
ALaSegunda

Test Plan

Version 1.3

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Test plan	Fecha: 14/12/2025
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Revisions

Date	Version	Description	Author
14/12/2025	1.3	UIT for Select Products, Purchase Products and Details	Héctor
13/12/2025	1.2	UI for Purchase Details	Héctor
13/12/2025	1.1	UI for Select Products & Create Purchase	Héctor
11/12/2025	1.0	Fix Product Price DTO & Purchase Constructors	Héctor

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Test plan

To prescribe the scope, approach, resources, and schedule of the testing activities. To identify the items being tested, the features to be tested, the testing tasks to be performed, the personnel responsible for each task, and the risks associated with this plan.

Test plan identifier

Identifier: ISII-2526-GrupoI-ALaSegunda-TP-UC2-Purchase-2025-12-14

Version: 1.3

Date: 14/12/2025

Scope: UC2 - Purchase

1. Introduction

2.1 Summary of Software Items and Features

The purpose of this test plan is to validate the functionality, usability, and stability of the Purchase Management Module and its associated User Interface (UI). The testing scope specifically covers the recent developments implemented in the Product Selection and Purchase Creation workflows.

Key features to be tested include:

Product Selection UI: Verification of the interface for selecting products prior to purchase (Ref: Task 172, Task 149).

Purchase Creation: Validation of the complete flow to create a new purchase, including form fields and backend integration.

Purchase Details View: Verification of the UI displaying the details of a confirmed purchase.

Data Integrity: Validation of Product DTOs (Data Transfer Objects) regarding price accuracy and Purchase DTO constructors.

2.2 Need and History These features were developed to complete the "Purchase" user story, moving from backend definition to a fully interactive frontend.

History: Development occurred between Dec 11 and Dec 14, addressing requirements from Task 172 and Task 149. Initial implementations of the DTOs were refined to include pricing logic (PR #166), followed by the UI implementation (PR #168, #170) and the finalization of automated UI Tests (UIT) (PR #173).

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2. Test items

3.1 Identification of Test Items

The following software items are submitted for testing. The build version corresponds to the commits merged as of Dec 14.

Test Item ID	Item Name / Feature	Version / Revision	Description
TI-001	Product Selection UI	Build 14.12 (PR #173, #168)	The interface component allowing users to browse and select products. Includes the specific UIT (UI Test) suite developed in Task 172.
TI-002	Purchase Creation Flow	Build 14.12 (PR #168)	The logic and UI forms required to convert selected products into a Purchase entity (Task 149).
TI-003	Purchase Details UI	Build 13.12 (PR #170)	The view component that renders the read-only details of a finalized purchase.
TI-004	Backend DTO Definitions	Build 11.12 (PR #166)	ProductDTO (Price field) and PurchaseDTO constructors ensuring correct data mapping between API and UI.

2.2 Transmittal Media & Requirements

- **Media:** The items are deployed in the [Development/Staging] environment via the repository's CI/CD pipeline.
- **Prerequisites:** Testing requires the latest backend API to be running and accessible.

2.3 References to Documentation

- **Requirements Specification:** See cards of github (<https://github.com/orgs/ISII2526GroupIALaSegunda/projects/1>) Task 172 and Task 149.
- **Design Specification:** See github pull requests #173, #170, #168.

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2.4 Exclusions

- **Delivery Assignment UI:** Testing of "Delivery Assignment" details (Task 164) is explicitly excluded from *this* specific test session, as it falls under a different ownership (Alejandro).

3. Features to be tested

The following features and combinations will be validated to ensure the UI behaves as expected and integrates correctly with the updated data structures.

3.1 UI: Select Products

- **Description:** Verify that users can browse and select products from the available list.
- **Test Design Ref:** Based on the scope of Task 172 (Develop UIT for select products).

3.2 UI: Create Purchase

- **Description:** Verify the complete flow of creating a purchase using the selected products.
- **Test Design Ref:** Based on the scope of Task 149 (Develop UI for select products and create purchase).

3.3 UI: Purchase Details View

- **Description:** Verify that the details of a purchase are rendered correctly in the UI after creation or selection.
- **Test Design Ref:** Based on the implementation in PR #170 (Develop UI for the details of a purchase).

3.4 Data Integration (DTOs)

- **Description:** Verify that the "Price" field is correctly handled in the Product DTO and that Purchase objects are instantiated correctly without errors.
- **Test Design Ref:** Unit tests covering the changes in PR #166 (Add Price to Product DTO, fix purchase DTO constructors).

4. Features not to be tested

The following features are explicitly excluded from this test plan as they fall under different development tasks or owners.

4.1 Delivery Assignment Details

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- **Reason for Exclusion:** This feature is developed under Task 164 (seeing details of a delivery assignment) and is managed by a different author (Alejandro). It is outside the scope of the Purchase UI workflow.

4.2 Backend POST Method Implementation

- **Reason for Exclusion:** While the UI interacts with it, the internal logic and unit testing of the creation of the POST Method are excluded.

4.3 Purchase Form IDs

- **Reason for Exclusion:** The specific task "Add ids to purchase form fields" by Héctor is considered a separate configuration task, unless it blocks the execution of the main UI tests.

5. Approach

5.1 Overall Strategy The testing approach adopts a bottom-up strategy, starting with data structure validation (DTOs) and moving to UI interaction and full workflow automation. The effort is divided into two major phases: Unit Testing for data objects and Automated User Interface Testing (UIT) for the visual workflows.

6.2 Testing Activities by Feature Group

- **Group A: Data Transfer Objects (DTOs)**
 - **Focus:** ProductDTO (Price field) and PurchaseDTO constructors (PR #166).
 - **Technique:** Unit Testing.
 - **Activity:** Verify that the PurchaseDTO constructors initialize objects correctly without null pointer exceptions and that the Price field in ProductDTO accepts and returns valid numerical values.
- **Group B: UI Components & Workflow (Purchase & Products)**
 - **Focus:** Select products list, Create Purchase form, and Purchase Details view (PR #168, PR #170).
 - **Technique:** Automated UI Testing (UIT) and Manual Visual Inspection.
 - **Activity:**
 - **UIT Execution:** Run the specific test suite developed in **Task 172** ("develop uit for select products").
 - **Validation:** Ensure the UI navigates correctly from product selection to the purchase creation form (Task 149).

5.3 Comprehensiveness & Completion Criteria

- **Code Coverage:** Unit tests for DTOs must achieve 100% constructor coverage (PR #166).
- **Requirement Coverage:** 100% of the acceptance criteria defined in Task 172 (UIT) and Task 149 (UI) must be mapped to at least one test case.
- **Traceability:** Test cases will be traced back to the specific Pull Request IDs (#173, #170, #168) to ensure every deployed change has a corresponding verification step.

5.4 Constraints

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- **Backend Dependency:** The UI testing for "Create Purchase" (Task 149) is strictly blocked by the availability and stability of the "POST Method" (PR #171) developed by JaimeSa.
- **Timeline:** All testing must be completed within the sprint window ending Dec 14, coordinating with the merge dates of the PRs.

6. Item pass/fail criteria

A test item will be considered **PASSED** only if it meets the following criteria:

6.1 Automated UIT Criteria (Task 172)

- The automated test suite ("develop uit for select products") runs to completion without errors or timeouts.
- The UI correctly identifies and interacts with the products list.

6.2 Functional Criteria (Task 149 & PR #170)

- **Create Purchase:** The system successfully generates a Purchase ID upon form submission (integrating with PR #168).
- **Details View:** The "Purchase Details" view (PR #170) accurately reflects the data entered during creation, including the correct product prices (derived from PR #166).

6.3 Data Criteria (PR #166)

- PurchaseDTO instantiation does not throw exceptions during runtime.
- The Price field in ProductDTO is correctly serialized/deserialized between the backend and the frontend.

6.4 Fail Condition

- Any "Blocker" or "Critical" severity bug found during the test execution of the "Select Product" or "Create Purchase" flows constitutes an immediate **FAIL** for the respective item.

7. Suspension criteria and resumption requirements

7.1 Suspension Criteria Testing activities on the Purchase UI and logic will be suspended immediately under the following conditions:

- **Backend API Failure:** If the backend "POST Method" returns consistent 500 Internal Server Errors or is unavailable, blocking the "Create Purchase" flow.

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- **Critical Blocking Defects:** Any defect that causes the application to crash upon opening the "Select Products" view (Task 172) or prevents the user from navigating to the "Purchase Details" view (PR #170).
- **Data Structure Mismatch:** If the updated ProductDTO (PR #166) fails to map the "Price" field correctly, rendering the UI test data invalid.

8.2 Resumption Requirements

- **Fix Verification:** Testing will resume only after the blocking issue is resolved and a new build is deployed to the test environment.
- **Regression Activity:**
 - If the suspension was due to API/DTO issues, the **Unit Tests** for DTO constructors (PR #166) must be re-run first.
 - If the suspension was due to UI crashes, the specific **UIT suite** (PR #173) must be re-executed from the beginning to ensure stability.

8. Test deliverables

The following documents and artifacts are to be delivered upon completion of the testing process:

8.1 Documents

- **Test Plan:** This current document outlining the strategy for the Purchase Module.
- **Test Design & Case Specifications:** The source code and scripts for the Automated UI Tests (UIT) developed under Task 172.
- **Test Logs:** Execution logs generated by the CI/CD pipeline during the run of PR #173 (UIT) and PR #166 (Unit Tests).
- **Test Incident Reports:** GitHub Issues created for any bugs found during the manual inspection of the "Purchase Details" UI (PR #170).

8.2 Data & Tools

- **Test Input Data:** JSON datasets used to mock the "Select Products" list and "Price" values for the DTOs.
- **Test Tools:** The specific test automation framework setup used to execute the UIT for select products (Task 172).

9. Testing tasks

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The following tasks are required to prepare for and perform the testing of the Purchase features.

Task 1: Environment Preparation

- **Description:** Ensure the test environment has the latest backend code deployed.

Task 2: Data Layer Verification

- **Description:** Verify that the ProductDTO includes the Price field and PurchaseDTO constructors are valid.
- **Resources:** PR #166 (Add Price to Product DTO).
- **Date:** Completed by Dec 11.

Task 3: UI Functionality Testing (Manual/Exploratory)

- **Description:** Manually verify the "Purchase Details" view and the "Create Purchase" form to ensure visual correctness.
- **Resources:** PR #170 (UI for details) and PR #168 (UI for select products and create purchase).
- **Dependencies:** Task 2 must be complete.

Task 4: Automated UIT Development & Execution

- **Description:** Develop and run the automated UI tests for the product selection workflow.
- **Resources:** Task 172 (Develop UIT for select products - PR #173).
- **Skills:** UI Automation scripting.
- **Date:** Finalized by Dec 14.

Task 5: Reporting

- **Description:** Compile pass/fail results from the UIT and close the associated Pull Requests.
- **Dependency:** Task 4 must be successful.

10. Environmental needs

Test environment: Backend API deployed and reachable (PR #171), frontend build including PR #168, #170, #173, and Chrome/Chromium for UIT execution.

Tools: Unit test runner used by the repository (DTO validation - PR #166), UIT framework configured in the project (Task 172), and CI/CD logs for evidence.

Test data and access: Seeded products list with valid Price values and a test user account allowed to create and view purchases.

11. Responsibilities

The following groups and individuals are responsible for the management, execution, and support of the testing activities defined in this plan:

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11.1 Management & Design

- **Test Lead / Designer:** Héctor is responsible for designing the Test Cases, creating the Test Plan, and developing the Automated UI Tests (UIT) for the Select Products workflow (Task 172).

11.2 Execution & Resolution

- **Test Executor:** Héctor will execute the automated UIT suite and perform manual validation of the Purchase UI (Task 149).

11.3 Support & Environment

- **Configuration Support:** Héctor is responsible for ensuring the test environment is prepared with the necessary HTML IDs on form fields to facilitate automation.
- **Excluded Items Owner:** Alejandro is responsible for the Delivery Assignment module, ensuring its development does not negatively impact the shared environment.

12. Staffing and training needs

12.1 Staffing Requirements

- **Test Engineer:** Required skill level includes proficiency in Java (for DTOs) and UI Automation frameworks (for Task 172 UIT). This role is currently filled by the feature developer (Héctor).

12.2 Training Options

- While the staff is familiar with the general project architecture, specific training was required for the Automated UI Testing (UIT) framework. The Test Engineer (Héctor) dedicated a self-study period to acquire the necessary skills to implement the automation scripts for the "Select Products" workflow (Task 172), ensuring the successful execution of the test suite.

13. Schedule

The testing schedule is aligned with the development sprint activities finalized between December 11 and December 14.

14.1 Test Milestones & Task Duration

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Milestone / Task	Start Date	End Date	Resource / Owner	Description
M1: Data Layer Validation	11/12/2025	11/12/2025	Héctor	Validation of Product DTO prices and Purchase constructors (PR #166).
M2: UI Core Implementation	12/12/2025	12/12/2025	Héctor	Manual test of "Select Products & Create Purchase" flow (PR #168).
M3: UI Details Validation	13/12/2025	13/12/2025	Héctor	Verification of the "Purchase Details" view rendering (PR #170).
M4: Automated UIT Finalization	14/12/2025	14/12/2025	Héctor	Execution and sign-off of the "UIT for select products" suite (PR #173).

14. Risks and contingencies

The following high-risk assumptions have been identified for this test plan:

14.1 Risk: Missing UI Identifiers

- **Description:** The automated tests (Task 172) rely on specific IDs in the HTML. The task "Add ids to purchase form fields" (Héctor) appears at the top of the list, potentially overlapping with test execution.

14.2 Risk: Scope Overlap

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- **Description:** Simultaneous modification of Product DTOs (PR #166) might affect the "Delivery Assignment" tests (Task 164) running in parallel.

15. Approvals

The following persons must approve this plan before the final release of the software version.

Name	Title	Signature	Date
Héctor Ruiz	Lead Developer / Test Author	_____	14/12/2025
Alejandro Gómez	Project Manager	_____	__/__/__
Elena María	Project Supervisor	_____	__/__/__